CHEMICAL REACTIONS SUMMARY WORKSHEET

- Atoms with partly filled electron shells tend to be unstable and reactive.
- Chemical reactions take place when atoms lose, gain or share electrons.
- Atoms bond together because of the attraction between opposite charges.

Common Chemical Reactions

1. *The more reactive metals react with water.*

General reaction: metal + water → hydrogen gas + metal hydroxide

Example: Calcium + water → hydrogen gas + calcium hydroxide

$Ca + 2\ H_2O \rightarrow H_2 + Ca(OH)_2$

(a) Sodium + water →

(b) $Na + H_2O$ →

2. *Many metals react with acids*.

General reaction: metal + acid → hydrogen gas + metal salt

Example: magnesium + hydrochloric acid → hydrogen gas + magnesium chloride

$Mg + 2\ HCl \rightarrow H_2 + MgCl_2$

(a) zinc + hydrochloric acid →

(b) + →

(c) calcium + sulfuric acid →

(d) $Ca + H_2SO_4$ →

3. *Most metals react with oxygen*. This can be a fast reaction (combustion) or a slow reaction (tarnishing).

General reaction: metal + oxygen → metal oxide

Example: magnesium + oxygen → magnesium oxide

$2\ Mg + O_2 \rightarrow 2\ MgO$

(a) copper + oxygen →

(b) Cu + O_2 →

4. *Acid and base reactions.*
   - Acids contain hydrogen ions (H^+) when dissolved in water.
   - Bases neutralise acidic solutions by removing H^+ ions.
   - Bases that are soluble in water are called alkalis. Alkaline solutions contain hydroxide ions (OH^-).

General reaction : acid + base → water + salt

Example: hydrochloric acid + sodium hydroxide → water + sodium chloride

$HCl + NaOH \rightarrow H_2O + NaCl$

(a) sulfuric acid + sodium hydroxide → water + sodium _____

(b) $H_2SO_4 + NaOH$ →

- *Metal hydroxides and metal oxides are bases.*

General reactions: Metal oxide + acid → water + salt

Metal hydroxide + acid → water + salt

Example: magnesium oxide + hydrochloric acid → water + magnesium chloride

$$MgO + 2\ HCl \longrightarrow H_2O + Mg(Cl)_2$$

(c) calcium hydroxide + hydrochloric acid →

(d) $Ca(OH)_2 + HCl \longrightarrow$

(e) calcium hydroxide + sulfuric acid →

(f) $Ca(OH)_2 + H_2SO_4 \longrightarrow$

(g) magnesium hydroxide + hydrochloric acid →

(h) $Mg(OH)_2 + 2\ HCl \longrightarrow$

5. *Metal carbonate reactions.* (Carbonates are basic because they react with acids)

General reaction: carbonate + acid → carbon dioxide + _____ + _____

Example:

calcium carbonate + hydrochloric acid → carbon dioxide + water + calcium chloride

$$CaCO_3 + 2\ HCl \longrightarrow CO_2 + H_2O + Ca(Cl)_2$$

(a) magnesium carbonate + hydrochloric acid → carbon dioxide + water + _____

(b) $MgCO_3 + 2\ HCl \longrightarrow$

(c) copper carbonate + sulfuric acid → carbon dioxide + water + _____

(d) $CuCO_3 + H_2SO_4 \longrightarrow$

(e) sodium carbonate + sulfuric acid →

(f) $Na_2CO_3 + H_2SO_4 \longrightarrow$

General reaction: metal carbonate $\xrightarrow{heat}$ carbon dioxide + metal oxide

Example: calcium carbonate $\xrightarrow{heat}$ carbon dioxide + calcium oxide

$$CaCO_3 \xrightarrow{heat} CO_2 + CaO$$

(g) magnesium carbonate $\xrightarrow{heat}$

(h) $MgCO_3 \xrightarrow{heat}$